

Culture consumption and social capital*

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Abstract

Introduction

Prevailing models of taste formation, transmission and decay in the sociology of culture posit that current tastes are largely a product of the individual's current network configuration. The standard assumption is that tastes are transmitted through existing network ties, in particular those social connections that are characterized by a high degree of sociodemographic similarity—"status homophily" (Lazarsfeld and Merton 1954)—between ego and alter (McPherson, Smith-Lovin, and Cook 2001). From this perspective, birds of a feather not only flock together, but largely due to this mutual flocking they end up singing, going to the movies, and reading together (Mark 1998b, 2003). A key explanatory strength of the network model is its ability to account for the fact that the distribution of tastes across persons in socio-demographic space is "lumpy" and clustered in proximal regions of this space. The model explains this by suggesting that the dynamics of cultural taste formation and transmission is best envisioned as a "local bandwagon" process of network-based imitation (Mark 2003). Individuals are construed as highly susceptible to the influence of their current contacts and by implication tastes are seen not as enduring aspects of the person but as malleable, "thin" contents liable to change when those network ties in turn change in their turn.

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And change they do. Recent longitudinal network research has shown that rather than being stable over time personal networks as constantly shifting with new contacts constantly being added and deleted throughout the adult life-course (Burt 2000, 2002; Morgan, Neal, and Carde 1997; Suitor, Wellman, and Morgan 1997; Wellman, Wong, Tindall, and Nazer 1997). Thus, what is truly surprising about over-time network “stability” over time is “how *unstable* intimacy is” (Wellman *et al.* 1997: 47). The empirical fact of network volatility coupled with the network assumption of taste as being largely a function of the cultural choices of the individual’s current set of contacts, implies large-scale volatility of those very same cultural tastes. Thus if we take the network perspective seriously, we should expect that as friends change, so do tastes, so we should expect small to moderate correlations between an individual’s taste at time t and her taste at time $t+n$.

Very different assumptions are at core of the cultural capital model of taste formation and acquisition (Bourdieu 1984; Holt 1998; Lizardo 2006). Instead of positing tastes as highly malleable contents at the perennial mercy of a constantly shifting network configuration, cultural tastes are conceived as highly *durable* dispositions toward broad types of cultural objects and performances (Frith 1998). These dispositions are formed in the family and school environment throughout childhood and early to middle adolescence (Smith 1995). Once set, they are highly resistant to change. The reason for this is that as Holt (1997) notes following Bourdieu (1990), tastes are part the person’s *embodied cultural capital* and thus represents a highly resilient aspect of the individual’s relationship to cultural objects (Bourdieu 1984), encoded in the memory traces, semi-conscious affective tendencies and habitual dispositions responsible for practical action (Bourdieu 2000).

Contrary to the network account, according to the cultural capital model tastes should not exhibit over-time volatility, but instead show high inter-temporal correlation. Lizardo (2006) uses this stability assumption to suggest that tastes can in fact be seen as probable (reciprocal) *drivers* of network connections rather than being seen as exclusively *determined* by those connections. Using two-stage instrumental variables and matching methods on cross-sectional data, that study uncovers evidence consistent with this reciprocal effect. Engagement in a wide variety of cultural practices increases network size net of standard controls and after statistically correcting for simultaneity bias. This is in agreement with dynamic theories of cultural and network evolution, which rather than postulating a one-way avenue of causation going from networks to cultural tastes propose a reciprocal (over-time) two-way casual interaction between the two (Carley 1991; Mark 1998a). However, in the aforementioned study the stability hypothesis remains an empirically unverified background assumption. Furthermore to reliance on cross-sectional data, the casual effect of cultural taste on network maintenance or even the more widely accepted effect of network turnover on cultural taste loss, has not been decisively established.

In this paper, I use individual-level longitudinal panel data to (1) empirically

adjudicate between these divergent empirical scenarios regarding stability implied by the network and cultural capital models, and to establish (2) direct and unambiguous casual connections between network turnover and cultural taste loss and (3) high rates of cultural activity and the maintenance of high rates of social connectivity in a longitudinal data analysis framework. In the next section, I go on to develop the empirical implications of the network and cultural capital views on more detail. In the following section, I introduce the data and propose my analytic strategy. In the next section, I present the results. I close with a discussion section highlighting the implications of the results for current theoretical understanding of the relationship between cultural tastes and social networks.

Network versus cultural capital theories of cultural taste loss: some empirical implications

Network theory

Contemporary network theories of cultural taste and attitude formation and change (and by implication decay) suggest that the contemporary taste profile of an individual is largely a function of the tastes held by the other persons in her surrounding social network (Erickson 1982, 1988). The key idea is that social ties serves as the “conduits” through which new cultural practices are *transmitted* and by way of which individuals *learn* these new cultural practices from their social contacts (Carley 1991; Mark 1998a). This proposition is shared by both straightforward network accounts (i.e. Cardon and Granjon 2005; Erickson 1996; Kane 2004) and more ambitious ecological-network theories of cultural taste (Mark 1998a, 2003; McPherson 2004). Mark (2003: 323-324) for instance speaks of social ties to friends and family as the most important sources of cultural taste acquisition: “[p]eople’s friends and family...influence the leisure activities they participate in...participation in various sports, visiting art museums, and gardening are...leisure activities that people often learn from parents, siblings, and/or friends.”

Network theories of taste transmission implicitly presume a relatively stable “social structure” coupled to relatively unstable patterns of cultural taste, otherwise it would be hard to think how something that varies at a more rapid pace “causes” or “affects” something that is more stable than itself (Simon 1952). It bears mention that this relative stability assumption is shared by the entire family of network theories that conceive of social structure as the pipes through which contents, or information “flows” and diffuses from one actor to another (Borgatti 2005). This implicit (and crucial) assumption of the stability of social structure, when this term is defined—as it is usually done in network theory—as the concrete patterns of network relations that bind individuals to one another (Berkowitz 1982, 1988; Wellman 1983, 1988; Wellman and Berkowitz 1988), can certainly be questioned in light of recent longitudinal research on constancy of network relations over time. This research—see the 1997 and 2005 special issues

of *Social Networks* (volume 19, number 1 and volume 27 number 4) and Burt 2000, for a thorough review)—shows that instead of being constant, network relations are extremely volatile, even when the time frame is constrained to relatively short periods of time, such as a year or less (Bidart and Degenne 2005; Suitor, Wellman, and Morgan 1997).

Most longitudinal studies of personal networks reveal surprisingly high levels of instability in personal networks over time (Lang 2000), with anywhere approximately about 30 to 60% of the personal network experiencing turnover in about a year's time. In his own research on bankers, Burt (2000: 11) finds that “three in four of the 22,709 colleague relations at risk of decay...are gone next year.” This means that old ties are constantly being deleted—what Burt (2000) refers to as “tie decay”—and new ones formed constantly throughout the life-course (Suitor, Wellman, and Morgan 1997; Wellman *et al.* 1997). Suitor and Keeton (1997: 57), report that more than half of the associates...named at T1 were not named again at either T2 or T3.” Morgan et al (1997: 14) conclude that their results “point to a considerable degree of instability in who is present in the network over time. In particular, the average overlap rate of 55% means that just over half of those who were named in one interview or the other appeared in both. Similarly, interpreting the average phi value of 0.34 as a test-retest correlation indicates only a limited ability to predict who was in the network from one time to the next.”

These findings imply that for most individuals a recurrent process of new contact selection, extant relationship “fine tuning” and possible deletion of old social ties is constantly at work (Lazarsfeld and Merton 1954). This represents an important—if ill understood—facet underlying the temporal evolution of a person's relational micro-environment. This dynamic acquires renewed salience in light of the observation that recently formed network connections appear to exhibit what Burt (2000), drawing on Stinchcombe (1965: 148-150), refers to as a “liability of newness.” This means that dyadic network ties tend to display a high probability of dissolution at early stages of the relationship, with the probability of termination steeply declining as the relationship matures. This proposition implies that the “tie decay function” (that is the probability of a tie being dissolved) is non-linearly decreasing in relationship age, with a high probability for young relationships which decreases as these dyadic ties ripen into close friendships. Burt (2000, 2002) finds evidence consistent with this hypothesis for both strong ties and network bridges (weak ties), respectively.

It is true that network stability is much more substantial in the “core” personal network, with the bulk of the instability concentrated on the periphery (Bidart and Lavenu 2005; Suitor, Wellman, and Morgan 1997; Wellman *et al.* 1997). However, most longitudinal network studies show that the stable “core” network tends to be composed of a small set of alters, primarily kin. The periphery of the network includes most of the persons usually referred to as (non-kin) “friends.” Studies show that there is considerable turnover even among those ties usually considered “close friends” (Bidart and Lavenu 2005). It is possible to

revise network theories of taste transmission to suggest that taste are mostly transmitted through the small, relatively stable core and not through the larger, unstable periphery. However, most contemporary network theories of taste do not make this distinction, and assume that taste are as equally likely to be learned from family and friends (Mark 2003). It is clear from recent studies on personal communities that the “core” kin network while mobilized for many purposes, including emotional and financial support, is not used for companionship and sociability (Simmel 1949) in any meaningful sense. Most research and theory on culture consumption communities shows that if tastes are exploited in interaction, this is done precisely with those contacts that are usually named as “friends” and which are seen outside the home and the neighborhood (DiMaggio 1987; Fiske 1987; Frith 1998). However, there is nothing inherent in the logic of the network theory of taste transmission that makes it incompatible with current data suggesting large rates of network turnover and the liability of newness of social ties.

Nevertheless, it is important to note that the social transmission hypothesis coupled with the empirical observation of the instability of social ties, jointly imply the inter-temporal instability of cultural tastes. The reason for this is that as Mark (2003) notes a taste that is not “reinforced” by social network influence through interaction is likely to be “forgotten.” This is a key assumption of the network based model of taste acquisition. Thus, if we retain the network assumption that individual tastes are largely a function of the tastes displayed by the persons current network alters, and if we accept the fact that these alters are constantly change over the adult life course, then we should expect cultural tastes to themselves exhibit high volatility over time:

H1: We should observe low to moderate inter-temporal stability of cultural taste across individuals.

Network theory of course, does not only imply volatile tastes, but directly suggests that any observed amount of taste loss should be primarily associated with life-course events associated with network turnover (i.e. geographical mobility, marriage and divorce, entering and leaving the workforce, adding or dropping social interaction foci, etc.). Surprisingly, but primarily due to the paucity of longitudinal data of cultural taste, this most crucial of assumptions of the network-theoretic view of cultural taste loss has never been assessed in an unambiguous way, although cross-sectional analyses appear to be consistent with computer simulated predictive scenarios (Carley 1991; Mark 2003). The network hypothesis therefore implies that:

H2: Observed cultural taste loss should be primarily a function of life-course changes associated with network turnover.

Cultural capital theory

The cultural capital approach to taste formation provides a very different picture of the stability of tastes than that which can be gleaned from contemporary

network theory. Instead of implying that tastes are unstable, “shallow” contents liable to be replaced at the same pace that network relations are replaced, the cultural capital approach suggest that taste are “deep” *durable dispositions and trans-temporally stable forms of embodied cultural capital* (Bourdieu 1986; Holt 1998). It is the relative stability of tastes vis a vis network relations that allows them to function as resources to help *sustain* and *maintain* these network relations over time—primarily by serving as topical resources in conversational interaction rituals (DiMaggio 1987; Lizardo 2006) providing a common focus of attention and affective commitment (Collins 2004). This helps to shift interactions away from purely “instrumental” purposes of information gathering and rational goal attainment to sociable interaction for its own sake (Granovetter 2002; Simmel 1949).

While there has been little empirical research on the dynamic stability of tastes through time, there is good reason to suppose that they are more stable than most network theory leads us to believe. For instance, most studies of the role of early family and school experiences on cultural capital have shown strong effects of arts participation, education and after-school training during adolescence on adult tastes even after controlling for subsequent educational attainment (Kracman 1996). Smith (1995) using data from the 1993 General Social Survey, shows that musical tastes are developed early in young adulthood and appear to be fairly stable across age-cohorts. Holbrook and Schindler (1989) find that persons tend to like styles of popular music with which they were familiar in late-adolescence over older or more contemporary styles, suggesting that tastes tend to be retained over the life-course. Bennett, Emisson and Frow (1999: 178), using qualitative evidence from retrospective interview reports on musical preferences in Australia find that “...many of those who were in middle age had retained the music preference of their late teens and early adulthood.” They conclude that while they “...cannot discount the possibility that some people will undergo radical changes in their tastes as they age, on balance the evidence appears to favor the explanation that the taste acquired in the formative years of a cohort persist into later life” (1999: 180). Crook (1997) finds strong correlations of reading and *beaux arts* practices on a three-year panel survey of the Australian population and also across self-reports of cultural activity in adolescence and adulthood, suggesting high stability of cultural practices across the life-course. From this perspective therefore, while the effect of social network change on cultural taste loss is not denied, whatever relational (or “network”) effects on taste exist, must operate in the background of high-levels of cross-temporal taste stability and the reciprocal action of cultural taste on social networks:

H3: We should observe high-levels inter-temporal stability of cultural taste across individuals.

According to the cultural-capital theory inspired (i.e. Bourdieu 1986) “conversion model” of cultural into social capital (DiMaggio and Mohr 1985; Dimaggio and Useem 1978; Lizardo 2006), the primary ways in which tastes are used to sustain social networks, is by allowing the individual to connect to alters that are

both relationally close—when specialist or “niche” cultural tastes are concerned—and relationally distant—when talking about aptitude for “popular” cultural forms. From this perspective, culture consumption and the ability to decode and appropriate symbolic goods produced in the urban, folk and “culture-industry” realms (Crane 1993) are important resources that individuals mobilize in the *construction* and *maintenance* of active social relationships (Cardon and Granjon 2005; DiMaggio 1987). In this respect the culture conversion model is consistent with Lazarsfeld and Merton’s (1954) process model of friendship formation, with similarity based on cultural taste and practices (DiMaggio 1987; Mark 2003) playing the role that “value-homophily” played in the original formulation.¹ As Cardon and Granjon (2005: 302) note in their recent study of the role of culture consumption in the formation of social connections among young adults in France, “. . . a large proportion of ordinary conversational interaction is generally devoted to comments. . . on cultural practices and recreational consumption. These are recurrent topics, present in a wide range of social situations. . . Many social relations are constructed, in different ways, around cultural and recreational activities.”

That the origins of new—and reproduction of previously existing—network connections comes from the deployment of cultural knowledge in interaction is implicit in previous models of the relationship between culture and social structure (Fine 1979). For instance, the Carley-Mark (Carley 1991, 1995; Mark 1998a, 2003) “constructural” model can be interpreted as an elementary schema of how culture can be translated into social connections, and how social structure (the distribution of chances to interact across persons in the system) and cultural structure (the distribution of cultural forms across persons) can be defined in an interdependent manner. The constructural model is based on a simple assumption of similarity (homophily), which postulates that the likelihood of a social tie increases with the cultural similarity between any given dyad (a dynamic process similar to the one proposed in Homans 1950 and Lazarsfeld and Merton 1954). In this way the probability that two persons will interact is driven by their cultural similarity. Interaction, in this positive feedback loop, in its turn increases cultural similarity, as individuals exchange their stocks of knowledge with one another.

Thus we should expect that individuals who command a wide variety of tastes which cut across these well defined boundaries (i.e. those who have been referred to as “omnivores” in recent research [Peterson 1992, 1997, 2005]) and who, given the stability assumption, will also be able to “keep” that “tolerant” taste pattern active over time—will also be more effective in sustaining high rates of social

¹Contemporary network models of taste formation through social network ties (Mark 1998b) can be seen as partially based on the Lazarsfeld-Merton formulation. However, these models raise what was explicitly a secondary dynamic in the original scheme—i.e. mutual “value-adjustment” through social interaction of initially different value dispositions (Lazarsfeld and Merton 1954: 33)—to being the only operative, and thus primary dynamic. It is clear however that for Lazarsfeld and Merton “selective” processes whereby contacts are sifted based on similarity of attitudinal dispositions similar to those emphasized in the culture-conversion account were considered primary (1954: 22-29).

interaction in their—constantly shifting, sparse and loosely bounded—personal communities (Wellman 1979; Wellman, Carrington, and Hall 1988; Wellman and Wortley 1990). This should also help the individual in maintaining ties to important “interaction foci” (Feld 1981, 1982) where old relations are maintained and new social connections are forged.

H4: High rates of social connectedness at a given time are directly related to cultural taste variety in the previous time period net of reciprocal influences of social connectedness on taste.

Data and variables

The data for this study come from the *Project Canada Survey* (PCS). The PCS is a longitudinal study of the social attitudes and cultural practices of a representative sample of Canadian citizens extending from 1975-1995, conducted out of the University of Lethbridge. The project began data collection in 1975 with a total 1,917 respondents. Every five years since representative samples of similar size have been collected, with a “core” set of respondents that have participated in previous surveys. Because it collected data on cultural practices and patterns of social interaction and membership in important foci of social interaction (i.e. voluntary association memberships) over time for the same individuals, the PCS data provide an excellent opportunity to deal with issues of casual order in the relationship between cultural taste loss and life-course events associated with network turnover.

In the following study, I use the two-wave panel covering the years 1985 and 1990. While there is data available for these same respondents for the year 1995, I restrict myself to the core set of respondents who participated in both of these waves (N=560).² I do this for two reasons. First, it is well known that cultural participation declines with age (Smith 1995), so it is advantageous to measure cultural taste with behavioral data before cultural activity begins to tail off. Second, these two waves used a comparable coding on the taste indicators (a slightly different coding was used in 1995) allowing an operationally valid measure of over-time taste loss.

Data limitations. While this data provide one of the few existing quantitative sources of information of cultural taste over-time for a large sample of respondents, there are some limitations of the data at hand that deserve mention: first, there are no direct measures of network connectivity (i.e. size of the personal network). The two measures available, frequency of interaction with friends, and number of associational memberships are important components of being sociable but certainly do not represent the notion of social connectivity in its entirety. Second, there are slight changes in the coding of the cultural activity and friendship

²This subset of the sample while not representative of the Canadian population can be used to monitor change in habits and practices and the determinants of these changes as it is attempted in the present study. Response rates for 1985 and 1995 were about 60%. About 65% of those interviewed in 1985 were available to be re-interviewed in 1995.

interaction variables across waves, with the introduction of a slightly more elaborate scheme in 1990 (including a “monthly” category) and changing to never category to a composite “hardly ever/never.” This may compromise inferences made in investigating taste-loss, by leading to an over-estimation of taste loss events for instance.

I attempted to deal with this last issue by counting a taste loss event as one that involved a shift from one of the extreme categories to the other (the coding of which did not change across waves) and by not using items that showed a clear effect of the shift in wording. For instance, it is clear that for the item related to “going to the movies” something like this happened, with 15% of respondents saying that they “never” go to the movies in 1985, but with 84% saying that they either “hardly ever or never” go to the movies in 1990. A cross-classification of the responses to the same question across time, reveals that this dramatic shift is mostly due to the fact that most of the respondents who answer “seldom” in 1985 (55%), were overwhelmingly more likely to answer “hardly ever or never” in 1990. The same artificial taste shift occurs for other cultural taste indicators associated with *attending events outside the home* (going to a sports event, going to see a dramatic performance in a theater). Cultural practices that do not involve going outside of the home were shown to not be affected by the change in wording, and are therefore the items that I use below in the taste stability and taste loss analyses.

Variables

Cultural Taste indicators

Cultural taste loss indicators. The PCS collected information for the years 1985 and 1990 related to the frequency of consumption of a variety of leisure culture activities and mass media outlets. To investigate the hypotheses related to cultural taste loss, I examine nine activities that encompass both at-home consumption and outside leisure (Alexander 2003; Crane 1993; DiMaggio 2000; van Eijck and van Rees 2000): (1) listening to music, (2) going to the movies, (3) attending a sporting event, (4) reading the newspaper, (5) reading books of fiction, (6) watching videos, (7) engaging in a hobby, (8) reading magazines, and (9) following a sports team. This broad selection captures a wide array of potential taste changes over time across different domains of cultural participation. Respondents were asked to report the frequency with which they engaged in each of these activities using the following prompt:³

Thinking now of your life outside of formal work, how often would you estimate that you: [i.e. listen to music]?

Four categories of responses were allowed: (1) very often, (2) sometimes, (3)

³The various codebooks for all waves of the PCS survey (1975, 1980, 1985, 1990 and 1995) are available online at: <http://www.thearda.com/Archive/IntSurveys.asp>.

Seldom and (4) Never.⁴ Thus for each respondent (i), for a given cultural activity (k) at time (t) is defined as their score in the respective ordered categorical variable y_{ikt} .

Cultural taste breadth indicators. To assess the hypotheses connected to the effects of cultural taste on indicators of sociability and social connectivity, I created a cultural taste breadth variable for the 1980, 1985 and 1990 waves. This is a simple additive index indicating participation in a wide range of cultural activities. Using the same set of items, the respondent receives one point in the cultural taste breadth scale if he or she reports engaging in the following cultural practices either “very often” or “sometimes”: (1) listening to music, (2) going to the movies, (3) going to a theatric performance, (4) going to a sports event outside the home, (5) watching videos at home, (6) engaging in a hobby, (7) reading magazines, (8) following a sports team, (9) reading a novel or book, and (10) reading the newspaper.

Network turnover indicators

As noted above, network theory and research provide several guidelines as to what life-course events are associated with radical changes in the personal network. The PCS contains two measures that tap more or less direct transformations of the personal network: (1) longitudinal changes in frequency of interaction with friends and (2) over-time changes in memberships across a variety of voluntary associations (McPherson 1983; McPherson and Smith-Lovin 1987). I also use five other indicators of personal network turnover based on positional shifts: (3) changes in educational status (Suitor and Keeton 1997) (4) changes in work status (Fischer and Olicker 1983), (5) changes in geographical location (Degenne and Lebeaux 2005), (6) changes in marital status (Fischer and Olicker 1983) and (7) changes in child rearing status (Munch, McPherson, and Smith-Lovin 1997). Since a non-negligible part of the individual’s personal network comes from contacts that are often seen at work (Brass 1985; Ibarra 1992; Straits 1996), we should expect that either entrance into or exit from the labor force, or shifts from part-time to full-time work, should have a clear impact on the size and composition of the personal network over time (Bidart and Lavenu 2005). The same can be said of geographical mobility (Relish 1997), with geographically mobile respondents more likely to be subject to radical transformation of their personal networks from those who are not mobile.⁵ Finally, a large body of research shows that family status transitions, such as getting married or getting divorced (Gerstel 1988; Leslie and Grady 1985; Milardo 1987; Pahl and Pevalin 2005) and experiencing widowhood (Anderson 1984; Morgan, Carder, and Neal

⁴As noted above, the 1990 coded the same variables using a slightly more elaborate categorization. I recoded those variables to match the 1985 scheme.

⁵The nominal employment status from which the binary indicator for changes in work status is constructed has six categories (1) full time, (2) part time, (3) unemployed, (4) retired, (5) keeping house, and (6) other. The geographical mobility item is based on the region in which respondent reported residing in each wave. The marital status variable has five categories: (1) married, (2) cohabiting, (3) divorced, (4) widowed and (5) never married.

1997; Morgan, Neal, and Carde 1997), have significant effects on the social networks of men and women. We should thus expect that transformations in family life, such as getting divorced, getting married or having children (Belsky and Rovine 1984; Bott 1971; Munch, McPherson, and Smith-Lovin 1997), should result in direct effects on the size and composition of the personal network.

Social connectedness indicators

In order to test the cultural capital hypotheses (3 and 4), I use two indicators of social connectivity: (a) the count of close friends with whom the respondent interacts, and (b) the ability to remain connected to important foci for social interaction, such as voluntary associations.

Retention of connections to important interaction foci is measured using a scale that simply counts the distinct types of voluntary associations the respondent belongs to in each wave of the survey: these include private clubs, sport clubs, service organizations, and hobby clubs. These types of memberships are the kinds of interaction foci in which mass media and arts-related culture-mediated social interaction is most likely to play a key role in the formation and maintenance of social contacts (Long 2003), and thus on the likelihood that the individual will retain or drop the membership (McPherson, Popielarz, and Drobnic 1992; Popielarz and McPherson 1995).

Sociodemographic controls

The models shown below include the following sociodemographic controls: gender, age, education, city size, employment status, marital status, and presence of children in the household. The descriptive statistics of the variables are detailed in Table 4 of the Appendix.

Network driven taste loss

Analytic Strategy

The cultural taste loss model can be thought of as a discrete-time event history analysis for two time periods (1985 and 1990) (Allison 1982). The dependent variable equals 1 if a cultural taste loss event is observed (the individual reported engaging in the activity “very often” in 1985, but “seldom” or “never” in 1990) and 0 otherwise.

To model this, I employ a pooled discrete-time logistic regression equation estimating the log odds of taste loss for each of the nine cultural activities. The predictors include a vector of network turnover indicators (e.g., changes in marital status, changes in work status, geographical mobility, changes in friends) as well as the standard matrix of sociodemographic controls. To account for the fact that observations for the same individual are not independent across time periods (unobserved heterogeneity within respondents), I estimate the models with cluster-robust standard errors (clustered by individual respondent). If

the network hypothesis is correct (hypothesis 2), we should observe significant, positive coefficients for the network turnover indicators, implying that relational disruptions drive taste loss.

Results

Frequency of Taste Loss. Consistent with the cultural capital hypothesis, across all nine cultural taste domains, taste loss is a relatively rare event. This is the case whether this is defined in a “strong” form (from “very often” in the first wave to “never” in the second) or in a weaker form (from “very often” to *either* “seldom” or “never”). Less than 1% of respondents go from having a strong taste in listening to music or reading the newspaper to losing that taste in the five-year period. A relatively larger number of respondents lose the taste for reading books (1.07%) or following sports (1.43%), but the main empirical finding here is that the great majority of respondents retain their tastes ($\approx 98\%$). When taste loss is defined in a weaker form, the worst retention rate drops to about 95% for reading books. This supports the cultural capital assumption that tastes are relatively stable over time, exhibiting stability rates that far exceed what is typically observed for personal networks (Suitor, Wellman, and Morgan 1997; Wellman *et al.* 1997).

Taste loss Models. The coefficient estimates of the discrete-time logistic regression models are shown in the Taste Change table below. All of the models have as the dependent variable a taste loss event defined according to the “weak” criterion above. Consistent with the expectations derived from the network-based taste transmission theory (Hypothesis 2), I find that—however rare—the cultural taste loss that can be observed is largely due to positional and social status changes associated with network turnover.

Thus, among the best predictors of cultural taste loss are changes in the frequency of *social interaction with friends* and changes in the *number of organizational memberships* ($p < 0.05$ or $p < 0.10$ for several cultural activities). Other important determinants of taste loss are status transitions that have been shown in the literature to be clearly associated with network turnover, such as changes in educational standing and changes in geographic location. The strongest effects in terms of magnitude are in accord with the view that tastes change when the personal network is transformed.

Changes in work status also have the expected positive effects on taste loss for select domains (like videos), although they are not uniformly significant across the board. The weakest effect of the entire set of network turnover indicators on taste loss pertains to changes in family status. Changes in marital status is not a useful uniform predictor of taste loss once other sources of network turnover are held constant, and changes in the number of children in the household have little discernible uniform effect. Possible reasons for the paucity of effects of changes in marital status include measurement error due to underestimating family volatility or the fact that there is little net effect of family status changes

left after the primary relational outcomes of these changes (shifts in the frequency of interaction with friends) are held constant.

In all, it is safe to say that the network hypothesis is partially confirmed by these data. While the overall *rates* of cultural taste loss are so low as to conflict with standard network theory’s assumptions of high taste malleability, the *sources* of taste loss when it does happen are largely those that we would expect: transformation of the personal network due to life-course changes associated with network turnover.

Longitudinal Culture conversion model

Analytic Strategy

In this section, I estimate the potential causal effect of cultural capital (as measured by taste diversity) on the maintenance of network connectivity over time. Estimating this reciprocal effect longitudinally presents two major methodological challenges: unobserved individual heterogeneity (time-invariant individual-level factors correlated with both variables) and simultaneity bias.

To deal with these issues and properly model count-based network outcomes, I present a unified longitudinal modeling framework:

Mundlak (Within-Between) Poisson Models. I employ a Mundlak (within-between) specification using a Poisson regression framework. Standard random-effects models require the stringent assumption that unobserved individual-level heterogeneity is entirely uncorrelated with the regressors. The Mundlak approach relaxes this assumption by explicitly including the person-specific longitudinal means (the "between" effect) of the time-varying covariates alongside the group-mean-centered deviations (the "within" effect).

This allows us to cleanly disentangle the “within-person” effect (how temporary changes in an individual’s cultural taste over time affect changes in their social connectivity) from the “between-person” effect (how stable, average differences in cultural tastes between individuals relate to average differences in social connectivity), thereby minimizing omitted variable bias.

Furthermore, because the primary outcome variables for social connectivity (the number of friends and the number of voluntary association memberships) are discrete, non-negative integer counts rather than continuous normal variables, these models are estimated using Poisson regressions. This ensures that the estimates correctly respect the variance structure of the count data.

Results

The culture conversion model (Lizardo 2006) implies that individuals who have command of a wide variety of tastes will be able to sustain larger and sparser personal networks over time, allowing them to reap the benefits of those social connections in the form of higher than average rates of social interaction and a

Table 1: Logistic Regression of Taste Change (with Clustered SEs)

[illegible]

higher ability to remain connected to important social foci where new contacts are made and old ones are sustained.

The results of the Mundlak (within-between) Poisson models of the effect of cultural capital (in the form of “omnivore” taste breadth) on network outcomes across time periods are shown below.

The results are highly consistent with the culture conversion model, particularly when examining the robust between-person differences (the ‘mean’ effects). I find that persons who display a persistently wide variety of tastes (a higher longitudinal mean of cultural breadth) sustain higher frequencies of social interaction with friends ($p < 0.05$). While this cross-sectional “between” effect does not offer definitive evidence of strict causality, it cements in a robust longitudinal modeling framework the results previously reported by Lizardo (2006). Broad cultural tastes are strongly associated with sustaining larger and more active patterns of interpersonal engagement over time.

Furthermore, having a high average repertoire of tastes is not only useful in sustaining higher levels of informal social connectivity, but it also significantly raises the chances of remaining tied to important social foci of interaction, such as voluntary associations ($p < 0.05$). Because recruitment (and the over-time stability of membership) into these foci is largely a matter of keeping alive the network ties that bind the individual to other members of the group (Popielarz and McPherson 1995), it is no surprise that whatever resource helps the individual to sustain social connections to other persons will also appear to sustain his or her connection to these larger social entities.

The Mundlak specification also allows us to evaluate the within-person effects—whether a temporary spike in an individual’s cultural tastes over their own average translates to immediate gains in connectivity. The within-person effects for cultural taste breadth are smaller and only marginally significant for interaction with friends ($p < 0.1$) and non-significant for organizational memberships. This suggests that it is the durable, trait-like possession of cultural capital (the between-person effect), rather than short-term fluctuations in cultural activity, that primarily drives sustained social connectivity and organizational integration over the life course.

Discussion and conclusion

This paper advances current theory and research in several ways. First, it provides for the first time conclusive empirical evidence that pertains directly to the empirical adequacy of different assumptions regarding the question of the stability of cultural tastes over time. The question of taste stability is important, since it bears on the direct empirical implications of the two most dominant sociological theoretical frameworks that help us explain the relationship between social ties and cultural resources: network theory whether in its relational (Erickson 1996) or ecological (Mark 1998a, 199b; McPherson 2004) variants, and cultural capital theory (Bourdieu 1984, 1986). Second, this paper produces direct

Table 2: Mundlak (Within-Between) Poisson Models for Network Stability

	Friends (Mundlak Poisson)	Org. Memberships (Mundlak Poisson)
(Intercept)	0.823* (0.176)	-2.147* (0.363)
numcult_within	-0.039+ (0.022)	0.018 (0.035)
educ_within	-0.018 (0.047)	0.039 (0.072)
married_within	0.112 (0.184)	-0.287 (0.292)
childre_within	-0.053 (0.093)	-0.024 (0.151)
bigcity_within	0.175 (0.288)	-0.233 (0.471)
numcult_mean	-0.068* (0.016)	0.201* (0.032)
educ_mean	-0.010 (0.017)	0.089* (0.033)
married_mean	0.080 (0.066)	0.085 (0.130)
childre_mean	-0.012 (0.016)	0.022 (0.031)
bigcity_mean	-0.002 (0.042)	-0.056 (0.079)
female	0.082 (0.050)	0.329* (0.100)
white	-0.065 (0.127)	0.260 (0.268)
as.factor(wave)4	0.495* (0.073)	-0.033 (0.118)
SD (Intercept id)	0.000	0.476
Num.Obs.	1116	1116
R2 Marg.		0.114
R2 Cond.		0.316
AIC	3219.5	2636.3
BIC	3294.7 ₁₆	2711.6
ICC		0.2
RMSE	0.78	0.71

+ p < 0.1, * p < 0.05

empirical evidence of another crucial assumption of network-theoretic views of taste formation, transmission and decay: that of the effect of relational disruption on taste loss. I tackle this question by attempting to link longitudinal shifts in patterns of social interaction, connection to important social foci (Feld 1981) and life-course transitions associated with network turnover to changes in cultural taste. This represents the first time that this most crucial of implicit assumptions of network theories of taste (Mark 1998b) receives empirical treatment in a research design where issues of casual direction can be dealt with satisfactorily. Finally, this paper builds on previous research on the role of cultural capital and other types of cultural resources on the creation and maintenance of social ties (DiMaggio 1987, 1993; DiMaggio and Mohr 1985; Lizardo 2006; Mische 2003) by providing relatively unambiguous evidence of the reciprocal casual effect of durable patterns of cultural taste on the maintenance of patterns of social interaction and connection to social foci over-time.

The results show that the cultural taste instability corollary of traditional network accounts appears not to be operative in this data. Instead, taste for music, newspaper reading, books, and sports show high degrees of stability, a result consistent with the assumptions of a cultural-capital model. However, even though the taste-stability corollary is not supported by these data, the network theoretic assumption that cultural taste loss is more likely to be due to events associated with the disruption of personal networks receives very strong support. Across all nine cultural forms included in the analysis taste loss events are strongly associated with changes in interaction frequency with friends, alteration in patterns of connection to important social foci, and disruptions associated with geographic mobility and changes in work status. Finally, a more stringent test of the cultural conversion hypothesis than that offered by Lizardo (2006) produces robust substantiation for the theory. Even after taking into account both simultaneity and unobserved heterogeneity bias, we find that individuals who command a wide variety of cultural tastes are more likely to sustain higher levels of social interaction with friends and higher levels of connectivity to important interaction settings (such as voluntary association memberships) through the five year period. This effect it is important to underscore appears not to be due to the reciprocal effect of connectivity on cultural taste breadth.

The results presented here have significant implications for how we theorize the statics and dynamics of cultural taste formation, transmission and decay. Rather than the generic idea found in the traditional network model of the individual who is influenced by the tastes of those persons that he or she is connected to, the cultural capital derived hypotheses of relative taste stability and use of cultural taste for the formation of new relationships point to two important additional considerations usually ignored in traditional network-based accounts.

Choice-homophily of network contacts based on pre-existing cultural dispositions. Standard network theories of taste transmission draw on the notion of “homophily” (i.e. the empirical generalization that close friends tend to be similar in sociodemographic characteristics) to explain the fact that cultural tastes and

practices tend to cluster in certain areas of social space (Mark 2003). From the standard point of view homophilious social connections based on socio-demographic similarity *pre-exist* individual patterns of taste and this explains why cultural practices “lump” in certain areas of social space. That is, the logic of network-based theories leads to them to attempt to explain the fact that cultural practices occupy delimited niches in social space by pointing to two interrelated processes: (1) tastes tend to be associated with social background and (2) individuals tend to associate with those who are socio-demographically similar.

Yet, there is sufficient qualitative evidence (Cardon and Granjon 2005; Long 2003; Wali, Severson, and Longoni 2002)—and as this paper shows, quantitative evidence (see also Lizardo 2006)—to suggest that just in the very same that tastes are transmitted through pre-existing network ties, *new* social ties are formed *on the basis of pre-existing cultural dispositions* (Lazarsfeld and Merton 1954).⁶ This is the obverse of the traditional network-theoretic emphasis on the formation of new cultural tastes on the basis of preexisting social ties. In fact, given the relative stability of cultural tastes in comparison to network connections, it is likely that this last process may in fact occur more often than the former process (Lizardo 2006). In this respect (and in contradiction to standard network accounts), the “lumpiness” of cultural tastes in social space may be actually instead be the result of an active—but not necessarily conscious—process of contact selection and relationship fine-tuning *keyed around cultural compatibility* (Lazarsfeld and Merton 1954).

This proposition is concordant with Burt’s (2000) observation of the “liability of newness” of recently formed ties, and with recent studies of changes in social networks of young adolescents and young adults (Bidart and Degenne 2005; Bidart and Lavenu 2005; Cardon and Granjon 2005). Notice that if this is correct, then sociodemographic homophily could in the majority of cases be a *by-product* of cultural compatibility and not the other way around (this cultural compatibility may of course be itself due to socio-demographic of the family of origin [Bourdieu 1984; Illouz 1998]). This is an insight that was always implicit in the theoretical logic of dynamic simulations of network formation based on shared cultural knowledge (Carley 1991), but which has been occluded in recent theory and research due to the one sided emphasis put on the socio-structural determination of shared tastes.

Class-based heterogeneity in contact choice and structural determination of tastes. Another insight that is brought about by the current research is the fact that the extent to which contacts are subject to a process of active selection based on taste itself varies according to the individual’s command of cultural, educational and socio-economic resources. Put simply, high cultural capital individuals, in particular those employed in prestigious occupations that demand

⁶See Illouz 1998 on romantic partnerships and cultural capital DiMaggio 1996 and Ostrower 1998 on taste for the high arts and social relations among elites and Bennett 2004 on taste for popular music and social network formation in online communities.

of extensive degrees of cognitive and cultural skill not only have larger and sparser social networks (Lizardo 2006), but the proportion of their networks that is composed of constantly changing, loosely bounded social ties is larger and more heterogeneous than that of culturally disadvantaged individuals employed in routine, low-complexity occupations (Coser 1991; Fischer 1982). In this respect, the endogenous effect of cultural capital in both the ability to select among different social ties and the ability to *reach* across the social structure to form those ties, is itself partially determined by position in social space.

This is important, because the dominant way that “social space” has been conceptualized by network and organizational theorists (i.e. Hannan, Carroll, and Pólos 2003; McPherson 2004) in relation to culture has been—following Blau (1977)—as a multidimensional manifold composed of variables such as age and occupational prestige. The sociodemographic niche for a particular cultural or social form can then be given a metric definition in terms of these variables. Social distance among two points in these space is conceived as being proportional to sociometric distance in social networks, that as being a proxy for the probability that two individuals will interact in real life (Mark 1998a). This presumes that the socio-demographic covariates that define social space are strictly exogenous (i.e. they do not interact in the statistical sense) with the propensity to deploy cultural knowledge reaped from consuming the cultural form in question in interaction to form new social relationships with persons located at relatively distant locations in social space. Traditional network models also assume that there is no interaction between the occupancy of privileged locations in social space and the relative propensity to acquire or learn about specific cultural forms from proximate social contacts. This process is seen as being a generic dynamic applicable to all individuals regardless of their socio-demographic profile.

From the point of view of the culture-conversion model on the other hand the axes of social space as defined above are clearly composed of variables some of which (i.e. educational attainment) themselves are implicated in the extent to which tastes may be driven by social network dynamics. Thus, from this point of view, occupying a privileged position in social space plays an important role in the degree to which individuals are able to loosen the limiting effects of network-based social constraint. This happens due the greater ability of those who possess higher levels of educational and socio-economic resources to deploy a more heterogeneous repertoire of cultural knowledge in everyday interaction. This allows them to form and sustain social relationships that cut-across social dimensions of stratification and differentiation (DiMaggio 1987), as well permitting them to create and maintain intimate ties with those who share their knowledge and tastes. For this reason, the dimensions of “social space” cannot be considered strictly exogenous to the cultural contents whose sociodemographic niche they define. Instead differential command of these cultural contents on the part of individuals may themselves “warp” these dimensions, such that individuals located at some privileged locations in this manifold find it easier to traverse a large distance in that space and thus connect to a wide range of others (Erickson 1996; Lin and Dumin 1986). Individuals located in less privileged areas of the space, on the

other hand, are socially and culturally distant from most individuals and cannot connect very well across their locally bounded social circles (Bryson 1997).

There is plenty evidence that is consistent with these considerations. For instance Wellman (1979; Wellman et al 1988; Wellman and Wortley 1990) and Fischer (1982) show that as socio-economic and educational status increases, individuals are much more likely to have role-segmented, specialized relationships in which a clear differentiation is made between those ties designed for sociability and companionship (usually labeled as “friends”) and social ties that are the result of non-negotiable statuses such as kinship ties. These lines of research on personal community networks also show that individuals are much more likely to prefer to obtain companionship from voluntary rather than involuntary ties. These are the social contacts that the culture conversion model predicts are likely to be formed and sustained through pre-existing forms of cultural compatibility. These are also the social relationships that are likely to be sustained through interaction rituals in which the symbolic goods coming from artworlds and cultural industries figure as the primary subject of conversation (Cardon and Granjon 2005; DiMaggio 1987). Finally, these are also the same social ties that are continually undergoing high rates of turnover and being constantly recycled and reformed, and which thus require the recurrent deployment of cultural capital in interaction. This goes a long way toward explaining the almost insatiable demand for novel symbolic goods among high cultural capital class fractions (Douglas and Isherwood 1996).

In essence, for a relatively privileged cultural and economic elites, social connections to culturally affine others have come to be disembedded from place, kinship and ascription (DiMaggio and Mohr 1985; Holt 1998), and have taken the form of increasingly longitudinally unstable and thus constantly changing personal community networks (Wellman et al 1988). This increases the chances that differential command of the ability to consume a wide variety of cultural forms becomes a *driver* of the recurrent formation of new network relations. This is less likely to be the case for those individuals who are restricted to a more homogenous, kin-based, and locally constrained set of core alters who change relatively little over time. Consistent with this claim, Lizardo (2006) shows that after holding cultural capital constant, the positive effect of education and socio-economic status on social network size is reduced significantly. This has the curious implication that traditional network theory may more accurately describe the process of cultural taste formation among low cultural capital individuals subject to high levels of network constraint (Burt 2005) while the process of culture conversion may be a more accurate representation of the deployment and operation of cultural taste among culturally advantaged elites.

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Table 4. Means, standard deviations and range of the variables used in the analysis. Project Canada Survey 1985-1995

Variable	N	Mean	Std Dev.	Min	Max
Δ Social Interaction	561	0.81	--	0	1
Δ Organizational Memberships	561	0.79	--	0	1
Δ Educational Status	561	0.33	--	0	1
Δ Geographic Location	561	0.53	--	0	1
Δ Work Status	561	0.24	--	0	1
Δ Marital Status	561	0.08	--	0	1
Δ N. of Children	561	0.08	--	0	1
Marital Status in 1985 (Married=1)	561	0.83	--	0	1
Marital Status in 1990 (Married=1)	561	0.81	--	0	1
Work Status in 1985 (Employed=1)	561	0.65	--	0	1
Work Status in 1990 (Employed=1)	561	0.58	--	0	1
Gender (Women=1)	561	0.67	--	0	1
City Size in 1985 (Large City=1)	561	0.53	--	0	1
City Size in 1990 (Large City=1)	561	0.52	--	0	1
Education in 1985	561	2.71	1.30	0	5
Education in 1990	561	2.85	1.29	0	5
Age in 1985	561	51.53	13.30	19	87
Age in 1990	561	55.96	14.12	24	88
N. of Organizational Memberships in 1985	561	1.97	1.40	0	6
N. of Organizational Memberships in 1990	561	1.89	1.36	0	9
Freq. of Social Interaction in 1985	561	1.54	0.58	1	3

Variable	N	Mean	Std Dev.	Min	Max
Freq. of Social Interaction in 1990	561	2.78	1.02	1	5
Culture Consumption Scale in 1980	339	6.76	1.19	3	8
Culture Consumption Scale in 1985	560	6.43	1.72	1	10
Culture Consumption Scale in 1990	556	3.74	1.40	0	8

It is well known that measurement error attenuates the relationships between variables. This means that the relationship between two observed variables which are subject to measurement error will generally be weaker than their true relationship. For the analysis of change, this phenomenon implies that when the observed states are subject to measurement error, the strength of the relationships among the true states occupied at two subsequent points in time will be underestimated, or in other words, the amount of change will be overestimated. When the data are subject to measurement error, the observed transitions are, in fact, a mixture of true change and spurious change resulting from measurement error (Hagenaars, 1992; Van de Pol & De Leeuw, 1986). The latent Markov model makes it possible to separate true change and spurious change caused by measurement error (Vermunt et al, 1999: 184).

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Mixed Markov Latent Class Models

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Table 3: Logistic Regression of Taste Change (with Clustered SEs)

	music	movie	play	sports	paper	books	spevent	v
(Intercept)	−1.005 (0.912)	1.648 (1.989)	4.742 (2.908)	0.597 (0.943)	−0.733 (1.466)	−1.078 (0.933)	1.253 (1.270)	0
chnghfriendsTRUE	−0.126 (0.235)	0.284 (0.555)	0.775+ (0.457)	0.556* (0.247)	0.202 (0.291)	0.081 (0.230)	0.311 (0.329)	−
chorgs4	0.090 (0.126)	0.243 (0.232)	−0.438 (0.341)	−0.049 (0.135)	−0.057 (0.156)	0.128 (0.131)	0.217 (0.207)	0
chngheducTRUE	0.229 (0.192)	1.405* (0.617)	−0.231 (0.426)	0.188 (0.207)	0.201 (0.223)	−0.145 (0.195)	0.108 (0.316)	−
chnghmaritalTRUE	0.319 (0.344)	−1.555* (0.640)	−0.753 (0.660)	0.573 (0.427)	0.579 (0.382)	0.309 (0.333)	−0.572 (0.497)	−
chnghchildreTRUE	−0.089 (0.345)	−0.463 (0.546)	0.150 (0.835)	0.215 (0.376)	0.701+ (0.392)	−0.457 (0.358)	−0.835* (0.408)	−
chnghwrkstatTRUE	0.409+ (0.223)	0.059 (0.486)	−0.500 (0.445)	−0.005 (0.235)	−0.159 (0.267)	0.136 (0.227)	−0.045 (0.371)	−
chnghareanamTRUE	−0.039 (0.208)	0.267 (0.419)	−0.529 (0.480)	0.021 (0.219)	0.366 (0.249)	0.575* (0.215)	0.126 (0.334)	0
agegrp3	−0.098 (0.133)	0.355 (0.272)	−0.224 (0.326)	0.162 (0.138)	−0.473* (0.158)	−0.118 (0.136)	0.533* (0.195)	0
educ3	0.008 (0.076)	−0.147 (0.172)	0.100 (0.193)	0.042 (0.081)	−0.047 (0.094)	−0.218* (0.080)	−0.299* (0.130)	0
income3	0.019 (0.056)	−0.063 (0.141)	0.039 (0.124)	−0.001 (0.058)	−0.177* (0.069)	−0.002 (0.057)	0.083 (0.095)	−
married3	0.295 (0.330)	−0.931 (0.790)	−1.474 (0.945)	−0.213 (0.361)	−0.545 (0.366)	0.453 (0.314)	−0.901 (0.579)	−
childre3	0.023 (0.072)	0.105 (0.159)	0.265+ (0.160)	−0.083 (0.075)	0.097 (0.080)	−0.046 (0.073)	−0.118 (0.126)	−
working3	−0.130 (0.240)	−0.213 (0.542)	0.216 (0.567)	0.281 (0.250)	−0.033 (0.293)	0.438+ (0.240)	0.291 (0.370)	−
bigcity3	−0.362+ (0.210)	0.181 (0.426)	−0.943+ (0.496)	−0.013 (0.219)	−0.007 (0.245)	0.367+ (0.216)	−0.182 (0.323)	0
female	0.834* (0.257)	0.064 (0.507)	0.195 (0.675)	−1.121* (0.278)	0.401 (0.294)	0.811* (0.247)	−0.455 (0.437)	−
white	−0.015 (0.542)	0.480 (1.080)	0.488 (1.236)	−0.198 (0.603)	1.586 (1.167)	0.359 (0.567)	0.524 (0.705)	0
Num.Obs.	540	536	533	531	537	538	537	
AIC	747.7	246.1	224.5	705.9	580.3	716.8	357.8	0
BIC	820.7	318.9	297.3	778.6	653.2	789.7	430.6	7
Log.Lik.	−356.853	−106.027	−95.272	−335.959	−273.163	−341.403	−161.886	−3
RMSE	0.48	0.22	0.21	0.47	0.41	0.47	0.29	1
Std.Errors	by: id	by: id	by: id	by: id	by: id	by: id	by: id	1

Table 4: Mundlak (Within-Between) Poisson Models for Network Stability

	Friends (Mundlak Poisson)	Org. Memberships (Mundlak Poisson)
(Intercept)	0.823* (0.176)	-2.147* (0.363)
numcult_within	-0.039+ (0.022)	0.018 (0.035)
educ_within	-0.018 (0.047)	0.039 (0.072)
married_within	0.112 (0.184)	-0.287 (0.292)
childre_within	-0.053 (0.093)	-0.024 (0.151)
bigcity_within	0.175 (0.288)	-0.233 (0.471)
numcult_mean	-0.068* (0.016)	0.201* (0.032)
educ_mean	-0.010 (0.017)	0.089* (0.033)
married_mean	0.080 (0.066)	0.085 (0.130)
childre_mean	-0.012 (0.016)	0.022 (0.031)
bigcity_mean	-0.002 (0.042)	-0.056 (0.079)
female	0.082 (0.050)	0.329* (0.100)
white	-0.065 (0.127)	0.260 (0.268)
as.factor(wave)4	0.495* (0.073)	-0.033 (0.118)
SD (Intercept id)	0.000	0.476
Num.Obs.	1116	1116
R2 Marg.		0.114
R2 Cond.		0.316
AIC	3219.5	2636.3
BIC	3294.7 ₃₀	2711.6
ICC		0.2
RMSE	0.78	0.71

+ p < 0.1, * p < 0.05